GRADE 7

STEM OLYMPIAD

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

Sample paper

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is -3/4 + 5/8?
A. -1/8
B. 1/8
C. 3/8
D. 7/8

Q2. Solve: 3x + 7 = 4x - 5.
A. x=5
B. x=6
C. x=7
D. x=8

Q3. What is 15% of 360?
A. 45
B. 54
C. 60
D. 72

Q4. Simplify: 3a + 2a - a.
A. 2a
B. 3a
C. 4a
D. 5a

Q5. What is 2^5 x 2^3?
A. 2^8
B. 4^8
C. 2^15
D. 4^5

Q6. A circle has radius 7 cm. Using pi=22/7, its circumference is:
A. 22 cm
B. 44 cm
C. 77 cm
D. 154 cm

Q7. A rectangle is 15 cm by 8 cm. If each side is doubled, its area becomes:
A. 240 cm^2
B. 360 cm^2
C. 480 cm^2
D. 960 cm^2

Q8. The mean of 12, 16, 18, 22 and 32 is:
A. 18
B. 20
C. 22
D. 24

Q9. Which rational number lies between 1/2 and 3/4?
A. 2/5
B. 5/8
C. 4/5
D. 7/8

Q10. A product id first increased in price by 20%and then discounted by 20%. Compared with its original price, the final price.
A. The same
B. 4% higher
C. 4% lower
D. 8% lower

Q11. Pipe A fills a tank in 6 hours, while pipe B fills it in 12 hours. If both are opened together, how long will they take to fill the tank?
A. 3 hours
B. 5 hours
C. 4 hours
D. 6 hours

Q12.A 4×4 grid contains numbers from 1 to 16, each used exactly once. The sum of every row and every column must be equal. What must be the common sum?
A. 30
B. 32
C. 34
D. 36

Q13. An exterior angle of a triangle equals:
A. the sum of the two opposite interior angles
B. one interior angle only
C. 180° minus all angles
D. half the largest angle

Q14. The probability of getting an even number on a fair die is:
A. 1/6
B. 1/3
C. 1/2
D. 2/3

Q15. A car travels 150 km in 2.5 h. Its average speed is:
A. 50 km/h
B. 60 km/h
C. 65 km/h
D. 75 km/h

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Two identical plants are kept under identical conditions except for carbon dioxide concentration.
Plants A receives normal atmospheric CO_2
Plants B receives significantly more CO_2
After several hours, plant B initially shows a higher rate of photosynthesis.
A. Availability of light or water
B. Presence of gravity
C. Number of chromosomes
D. Length of the plant's roots only

Q17. In humans, most nutrients are absorbed in the:
A. stomach
B. small intestine
C. large intestine
D. mouth

Q19. Rusting of iron is a:
A. physical change
B. chemical change
C. change of state only
D. reversible change

Q20. A metal spoon is left in a cup of hot tea. After some time, its handle becomes hot. Which process is mainly responsible.
A. conduction
B. convection
C. radiation only
D. reflection

Sample Paper

Q18. A farmer's soil is found to be highly acidic. Which substance would generally be appropriate to help reduce the acidity?
A. Vinegar
B. Lime juice
C. Soaked lime
D. Carbonated water

Q21. Sea breeze occurs mainly because land and water:
A. heat at different rates
B. have the same temperature always
C. do not absorb sunlight
D. produce magnets

Q22. Which soil generally holds the most water?
A. Sandy
B. Clayey
C. Gravel
D. Rock only

Q23. A ring of bark is carefully removed from the stem of a tree without damaging the central woody region. After several weeks, swelling is observed above the ring. What is the best explanation?
A. Water accumulates above the ring because xylem is blocked
B. Food transported through phloem accumulates above the removed region.
C. Photosynthesis stops completely below the rings.
D. Roots begin producing extra food.

Q24. Respiration releases energy from:
A. food
B. soil
C. sunlight directly in animals
D. minerals only

Q25.A machine converts electrical energy into machinal energy with an efficiency of 75%. If it receives 800J of electrical energy, how much useful machinal energy does it produce?
A. 200J
B. 400J
C. 600J
D. 1,067J

Q26. An electromagnet becomes stronger when:
A. more current flows through more coil turns
B. the circuit is open
C. iron core is removed and current reduced
D. the wire is cut

Q27. The image formed by a plane mirror is:
A. virtual and same size
B. real and inverted
C. always smaller
D. always larger

Q28. Groundwater depletion is reduced by:
A. rainwater harvesting
B. paving all open land
C. over-pumping wells
D. cutting vegetation

Q29. A population of rabbits in a forest increases rapidly because predators become scarce. After some time, the rabbit population suddenly decreases. Which explanation is most likely?
A. Increased competition for limited food and resource
B. Rabbits stops requiring food after reproduction
C. Predators are necessary for rabbits to breathe
D. The forest produces unlimited food.

Q30. A gardener notices that plants produced from seeds show greater variation than plants produced through vegetative propagation. Why?
A. Seeds are always larger than vegetative parts
B. sexual reproduction involves genetic material from two parents, creating variation.
C. Vegetative propagation changes the DNA of every new plant.
D. Seeds contain no genetic information.

Sample paper

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A truss bridge is strong because triangles:

A. resist deformation
B. can never break
C. remove gravity
D. generate electricity

Q32. A robot has 4 possible instructions:
F=move forward 2 steps
B=move backward 1 step
R= turn right
L= turn left
It starts facing north at position (0,0).
It performs F→R→F→L→F→B.
What is its final position?

A. (2,3)
B. (2,4)
C. (1,3)
D. (1,4)

Q33. A student measures a pendulum period. For a fair test of length, which variable should be changed?

A. Length only
B. Length and mass together
C. Starting angle and length together
D. Clock used each time

Q34. A machine has input energy 500 J and useful output 350 J. Its efficiency is:

A. 35%
B. 50%
C. 70%
D. 85%

Q35. A bicycle gear with 20 teeth drives one with 40 teeth. The driven gear rotates:

A. half as fast
B. twice as fast
C. at the same speed
D. four times as fast

Q36. Which graph best shows distance changing continuously with time?

A. Line graph
B. Pie chart
C. Venn diagram
D. Tally chart only

Q37. A sensor gives readings 2, 5, 10, 17, 26…. Which number should come next?
A. 35
B. 36
C. 37
D. 38

Q38. A robot moves 6 m East, 4 m North, 2 m West, 1 m South. Its displacement from start is:

A. 4 m East and 3 m North
B. 8 m East and 5 m North
C. 4 m West and 3 m North
D. 6 m East and 1 m North

Q39. A solar cooker is improved by adding a reflector because it:

A. directs more solar radiation to the pot
B. increases friction
C. reduces sunlight
D. creates fuel

Q40. A substance X reacts with oxygen and produces a new substance Y. Y cannot easily be converted back into X by simple physical methods. Which statement is most appropriate?

A. X has undergone a chemical change
B. X has only changed its state
C. No new substance has formed
D. The process must be melting.

Sample paper

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A 2 kg object is raised 5 m. Taking g=10 N/kg, its gain in gravitational potential energy is:

A. 10 J
B. 25 J
C. 50 J
D. 100 J

Q42. A plant loses 30 mL water by transpiration in 2 h. At the same rate, loss in 5 h is:

A. 60 m
B. 75 mL
C. 90 mL
D. 150 mL

Q43. Which variable should be controlled when comparing how blade angle affects a wind turbine model?

A. Wind speed
B. Blade angle
C. Measured output
D. Research question

Q46. A mixture contains iron filings, sand and salt. What is the best sequence to separate all three?

A. Use magnet, dissolve salt in water, filter sand, evaporate water
B. Filter first, then magnet through water only
C. Evaporate first without adding water
D. Use only a sieve

Q47. A 1.5 kW heater runs for 2 h. Energy used is:

A. 0.75 kWh
B. 1.5 kWh
C. 3 kWh
D. 3000 kWh

Q48. A drone travels 120 m north then 50 m south. Its net northward displacement is:
A. 50 m
B. 70 m
C. 120 m
D. 170 m

Q44. An arch bridge carries load mainly by transferring force into:

A. compression along the arch
B. light
C. electric charge
D. sound

Q45. Which sequence is most scientific?

A. Question - hypothesis - test - analyse - conclude
B. Conclude - guess - test - ignore data
C. Build - decorate - conclude
D. Ask - answer without testing

Q49. In a design test, Prototype A holds 8 kg, B holds 11 kg and C holds 10 kg before failure. Which currently has the greatest load capacity?

A. A
B. B
C. C
D. All equal

Q50. A fair coin is tossed twice. Probability of two heads is:

A. 1/2
B. 1/3
C. 1/4
D. ¾

Sample paper